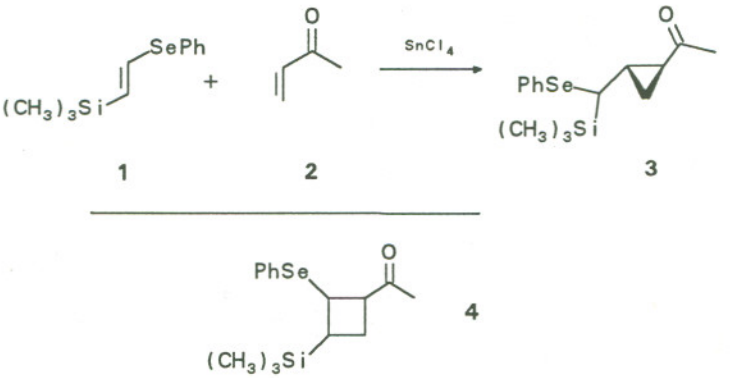
Organic Cume, February 15, 1992

Recently Yamazaki and coworkers[1] reported a new preparation for cyclopropanes using (E)-1-(phenylseleno)-2-(trimethylsilyl)ethene (compound **1**). When compound **1** is introduced to vinyl ketone **2**, product **3** was isolated in 42% yields (more details of the experiment are given in footnote 2).

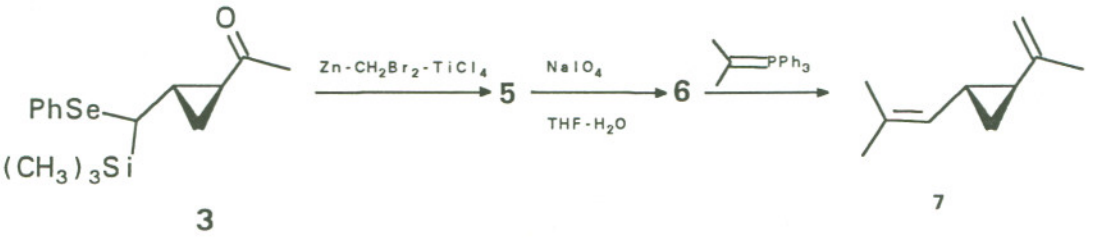
Yamazaki and coworkers **were surprised!!!** Before they carried out the reaction, they anticipated that the reaction between **1** and **2** would yield cyclobutane **4**.

A. Give a mechanism for the conversion of **1** to the anticipated (but not isolated) product **4**.



Compound **3** was converted to the natural product rothrockene **7** as shown below.

B. Give the structures of compounds **5** and **6**.



(1) Yamazaki, S.; Katoh, S.; Yamabe, S. *J. Org. Chem.* **1992**, *57*, 4-6.

(2) To a solution of $SnCl_4$ (366 mg, 1.40 mmol) in CH_2Cl_2 (1.8 mL) cooled to -78 °C, was added **1** (234 mg, 0.917 mmol) in CH_2Cl_2 (0.4 mL) followed by methyl vinyl ketone **2** (82.2 mg, 1.17 mmol). The mixture was stirred at -78 °C for 3 h. The reaction mixture was quenched by triethylamine (213 mg, 2.1 mmol), and then water was added to the mixture. The mixture was extracted with saturated $NaHCO_3$ and water, dried (Na_2SO_4) and evaporated in vacuo. The residue was purified by column chromatography over silica gel eluting with hexane-ether (4:1) to give **3** (184 mg, 62%).